

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.8: Vectors — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.8: Vectors
Driving Question	DRIVING QUESTION: Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Why Does 8 km Take You to Westlands or South B? Launching the Matatu Phenomenon	Two matatu routes from Kencom Bus Stop in Nairobi's CBD, both labelled 8 km, arriving at completely different destinations: Westlands (northwest) and South B (southeast). Road directional arrows and one-way signs encode direction in everyday infrastructure. A demonstration/video showed two people walking 8 steps in different directions and ending far apart.	A SCALAR quantity is described completely by magnitude (size) only — examples: temperature in Nairobi (22°C), time to cook ugali (20 min), distance run by Eliud Kipchoge (32 km). A VECTOR quantity requires both magnitude AND direction to be described completely — examples: displacement of the Likoni Ferry (mainland to island), velocity of a bodaboda heading south on Ngong Road, wind blowing from Lake Victoria eastward toward Kisumu Airport. The label '8 km' on a matatu route is a scalar — it tells us how far but not which way. A complete journey description is a vector.	The two matatus ended up at different destinations because 8 km describes only the magnitude (scalar) of the journey, not its direction. To distinguish Route A (8 km northwest → Westlands) from Route B (8 km southeast → South B), we need a mathematical object that carries BOTH magnitude AND direction — this object is called a VECTOR. Direction is already encoded in everyday road infrastructure (one-way signs, directional arrows), which means vectors are not abstract — they are built into the world around us in Nairobi's streets.
Lesson 2 Observe + Represent — Drawing the Matatu Journey as an Arrow	Learners observed that the two matatu journeys from Kencom — both 8 km long — produce two visibly distinct arrows on the Nairobi grid: one pointing north-west toward Westlands and one pointing south-east toward South B. They observed that moving a vector arrow to a new position on the Cartesian grid does not change the vector itself, as long as direction and length are preserved.	A vector is represented geometrically as a directed line segment (arrow) with a TAIL (starting point) and HEAD (arrowhead at the destination). Vectors can be written in bold ($\mathbf{v}$), underline ($\underline{v}$), column/component form, or AB-arrow notation. Two vectors are equivalent if and only if they have the same magnitude AND the same direction — position on the grid is irrelevant.	The phenomenon is now mathematically visible: the Kencom → Westlands journey-vector and the Kencom → South B journey-vector are drawn as two distinct arrows on the grid — different directions, different endpoints — even though both have magnitude 8 km. This explains concretely why the matatu drivers end up at different destinations: their journey vectors are not equivalent. The number '8 km' alone (a

			scalar) cannot distinguish them, but the directed arrow — a vector — can.
Lesson 3 Adding Vectors: Finding the Resultant of a Two-Leg Matatu Journey	On a squared-paper Nairobi grid, learners drew two successive displacement vectors — Kencom Bus Stop to University of Nairobi (vector a) and University of Nairobi to GPO (vector b) — and constructed the resultant vector using the triangle law (tip-to-tail), measuring its length and direction on the grid.	The resultant vector is the single displacement vector from the tail of the first vector to the tip of the last vector; it can be found geometrically using the triangle law (tip-to-tail) or the parallelogram law, and algebraically by adding the corresponding components of column vectors ($r = a + b$).	The reason two matatu drivers both travelling 8 km from Kencom CBD arrive at completely different destinations (Westlands vs South B) is that their displacement vectors point in different directions — the resultant of each driver's journey vectors, not the scalar total distance, determines the final destination.
Lesson 4 Scaling the Journey: Scalar Multiplication and Subtraction of Vectors	On a Nairobi-style 1 cm grid, learners drew four scaled versions of displacement vector $d = (2,1)^T$: the vectors $3d$, $\frac{1}{2}d$, $-d$, and $-2d$, starting from Kencom Bus Stop. They recorded each result in a table showing the column vector, magnitude, direction relative to d , and line of action.	Multiplying a vector by a scalar k changes its magnitude by the factor $ k $ but keeps it on the same line of action. A positive k preserves the direction; a negative k reverses it. Vector subtraction $a - b$ is the same as adding the negative vector: $a + (-b)$. These rules allow us to describe repeated, halved, or reversed boda-boda and matatu journeys precisely.	The results are explained by the definition of scalar multiplication: kd is a vector whose components are each multiplied by k , so its magnitude scales by $ k $ and its direction is determined by the sign of k . Subtraction works because $-b$ is a vector of the same magnitude as b but in the opposite direction, and adding it to a gives the vector from the tip of b to the tip of a .
Lesson 5 Column Vectors & Position Vectors — Giving Every Nairobi Landmark an Address	On a Nairobi coordinate grid with Kencom Bus Stop at the origin, we plotted position vectors $\vec{OA}$ to KICC at (2,3) and $\vec{OB}$ to Uhuru Park at (-1,2). We drew the arrows and placed the column vector notation beside each. We also looked at the two original matatu routes from Kencom — the Westlands route ending near (-6,5) and the South B route ending near (4,-6) — and wrote their column vectors side by side.	A column vector writes a displacement as a vertical pair of numbers: the top number gives horizontal movement (positive = east, negative = west) and the bottom number gives vertical movement (positive = north, negative = south). A position vector $\vec{OA}$ gives the location of point A relative to origin O. To find the vector from A to B we use $\vec{AB} = \vec{OB} - \vec{OA}$, subtracting the components. The two matatu drivers' journeys have completely different column vectors — (-6;5) versus (4;-6) — which is exactly why they arrived at different destinations even if their distances were similar.	Distance alone (a scalar) cannot distinguish two journeys that start at the same point — it tells you HOW FAR but not WHERE. A column vector captures BOTH the magnitude and direction of a displacement in a single algebraic object. Because the Westlands matatu moves predominantly west and north while the South B matatu moves east and south, their column vectors point in opposite directions. This is the mathematical reason two matatu drivers travelling the same distance from Kencom arrive at completely different destinations: their displacement column vectors are different, not just their distances.
Lesson 6 Magnitude & Midpoint — How Far Is the Fishing Ground, and Where Is the Halfway Mark?	Learners observed that the right-angled triangle formed by the horizontal and vertical components of a column vector has a hypotenuse whose length is exactly the straight-line displacement distance — and that applying Pythagoras' theorem to the two matatu column vectors from Lesson 5 gives magnitudes of approximately 7.8 km and 7.2 km, close to the '8 km' labels on the phenomenon map. Learners also observed,	The magnitude of a column vector $v = (x; y)$ is $ v = \sqrt{x^2 + y^2}$, derived from Pythagoras' theorem; this scalar value tells us how far a displacement is in a straight line but discards direction entirely. The midpoint M of the journey from point A($x_A; y_A$) to point B($x_B; y_B$) is $M = ((x_A + x_B)/2; (y_A + y_B)/2)$, giving the exact halfway position along the displacement. A negative component does not affect the magnitude	The '8 km' labels on the matatu map are the magnitudes of the two displacement vectors — they confirm that both drivers travel the same straight-line distance from Kencom Bus Stop. However, magnitude alone cannot determine destination: the Westlands vector (-6; 5) and the South B vector (4; -6) have similar magnitudes (~7.8 km and ~7.2 km) but point in completely different directions, producing

	on a coordinate grid, that the midpoint of a displacement vector from A to B lies at the average of the corresponding coordinates.	because squaring always produces a positive value.	completely different endpoints and different midpoints along the way. This proves that a scalar (8 km) is insufficient to describe a journey — you need the full vector, with both magnitude AND direction, which is exactly what the driving question demands.
Lesson 7 Translation Vector as a Transformation: Moving Shapes the Matatu Way	On the Cartesian plane, when triangle KNB was translated by vector (4 over 2), every vertex — K, N, and B — shifted by exactly the same vector arrow, landing at K', N', B'. Similarly, all four vertices of quadrilateral MJUA shifted by the same vector. Road arrows on the Nairobi street-grid handout all pointed in the same direction and represented the same column vector.	A translation moves every point of an object by the same translation vector, preserving shape and size but changing position. The translation vector can be found by subtracting the position vector of an object point from the position vector of its corresponding image point: $t = \text{image position vector} - \text{object position vector}$. Road directional arrows are real-world examples of translation vectors applied uniformly to all road users.	The two matatu drivers both travelled 8 km from Kencom Bus Stop but ended at different destinations because they followed different translation vectors — not just different distances. One driver's journey vector pointed north-west toward Westlands; the other pointed south-east toward South B. A scalar (distance = 8 km) cannot distinguish these journeys. Only a vector — capturing both magnitude and direction — completely describes where you will end up. Translation as a transformation makes this mathematically precise: every point of the matatu (or any object) is shifted by the same vector, which is why direction is the deciding factor in the final destination.
Lesson 8 Model Build, Consolidation & Final Explanation — Vectors in Nakuru: Synthesising the Complete Journey	Learners traced a three-leg boda-boda/matatu journey on a sketch map of Nakuru town (CBD Terminus → Westside Mall → ASK Showground → Lake Nakuru Viewpoint), colour-coding each leg and noting that the total path distance and the straight-line displacement from start to finish are not the same quantity.	Learners confirmed that each leg of a journey can be expressed as a column vector capturing both magnitude and direction; that column vectors are added component-wise to give the resultant displacement vector; that the magnitude of the resultant is found using Pythagoras' theorem; that the midpoint vector is the average of the start and end position vectors; and that the translation vector mapping start to finish is identical to the resultant displacement vector — together proving that scalar distance alone cannot locate a destination.	Learners wrote a Final DQB Answer resolving the anchoring phenomenon: the two matatu drivers from Kencom both travelled 8 km (equal scalars) but in different directions, producing different displacement vectors and therefore entirely different endpoints. A vector — encoding both magnitude and direction in column form — is the precise mathematical tool needed to describe a journey completely, predict a destination, and define the translation that maps one location to another.

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